

LAB 2

Question 1: Generic Class

Create a **Generic Class** named `DataStorage<T>` that can store and display values of different data types such as **integer, string, and double**. Create objects of the generic class for each data type and demonstrate that the same class can be reused for different types.

Question 2: Generic Method

Create a C# program that implements a **Generic Method** named `Swap<T>()` to exchange the values of two variables. Demonstrate the method by swapping two integers, two strings, and two double values, and display the values before and after swapping.

Question 3: Generic Collection

Develop a **Student Management System** using the generic collection `List<T>`. Store the details of at least five students, including **Student ID, Name, and Faculty**. Provide functionality to add new students, display all student records, search for a student by ID, and remove a student from the list.

Question 4: List Collection

Create a program using the `List<T>` collection to manage employee information. Allow the user to add employee records, display all employees, update an employee's salary, delete an employee record, and display the total number of employees stored in the collection.

Question 5: Dictionary Collection

Create a **Library Management System** using the `Dictionary<TKey, TValue>` collection. Store the **Book ID** as the key and the **Book Title** as the value. Implement features to add books, search for a book using its ID, update book information, remove a book, and display all available books.

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Question 6: Queue Collection

Develop a **Hospital Token Management System** using the **Queue<T>** collection. Add patients to the queue as they arrive, display the current waiting list, serve patients in the order they entered the queue, and display the next patient to be served.

Question 7: Stack Collection

Create a program that uses the **Stack<T>** collection to simulate the **Undo** feature of a text editor. Allow the user to enter multiple actions, undo the most recent action, display the current top action, and show all remaining actions in the stack.

Question 8: Comprehensive Programming Challenge

Develop a **College Management System** that demonstrates the use of **Generics and Collections**. Create a generic class to store different types of information, and use collections such as **List<T>**, **Dictionary<TKey, TValue>**, **Queue<T>**, and **Stack<T>** to manage students, teachers, library books, admission queues, and recent activities. Implement features to add, update, search, delete, and display records while demonstrating the practical use of each collection in a single C# Console Application.